

Global Food & Nutrition Dashboard

COMP5120 — Data Visualization

Nguyen Thi Tra My Tran Thi Hoai Phuong

VinUniversity

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Outline

- 1 Introduction & Motivation
- 2 Dataset Overview
- 3 Architecture & Techniques
- 4 Machine Learning Integration
- 5 Live Demo
- 6 Findings & Lessons Learnt

- 7 Q & A

Problem & Motivation

Core Question: What does the world eat, how healthy is it, and what's really in our food?

The Challenge:

- Food systems are incredibly complex and high-dimensional.
- We needed to bridge the gap between **macro-level** country calorie supply and **micro-level** product nutrition.
- Making massive scale data interpretable without losing the nuances of dietary composition.

Dataset Overview

Our automated data pipeline pulls from three major sources:

- **Our World in Data (OWID):**
 - 200+ countries, 60+ years of daily per capita caloric supply.
 - Dietary composition by food group.
- **World Health Organization (WHO):**
 - Adult obesity prevalence and child underweight data.
- **Open Food Facts:**
 - ~10,000 global products with macronutrients, Nutri-Score (A-E), and NOVA processing levels.

Architecture & Techniques

Tech Stack:

- **Framework:** Python Shiny (Reactive Programming)
- **Visualization:** Plotly, Matplotlib, Seaborn
- **Data Processing:** Pandas, NumPy
- **Machine Learning:** Scikit-learn

Reactive Design:

- Modular app structure for robust cross-filtering.
- Efficient state management to avoid redundant recomputations.

Machine Learning Integration

Rather than just displaying data, we embedded ML directly into the analytical workflow.

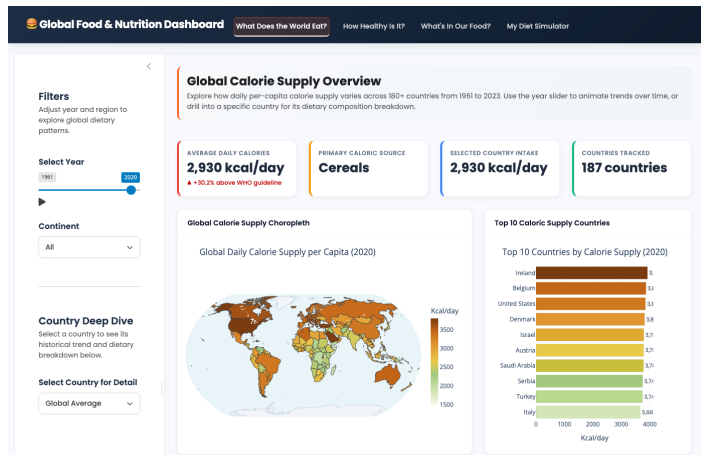
Unsupervised Learning (Country-Level)

K-Means Clustering: Profiles and groups countries based on their dietary composition to uncover hidden global patterns.

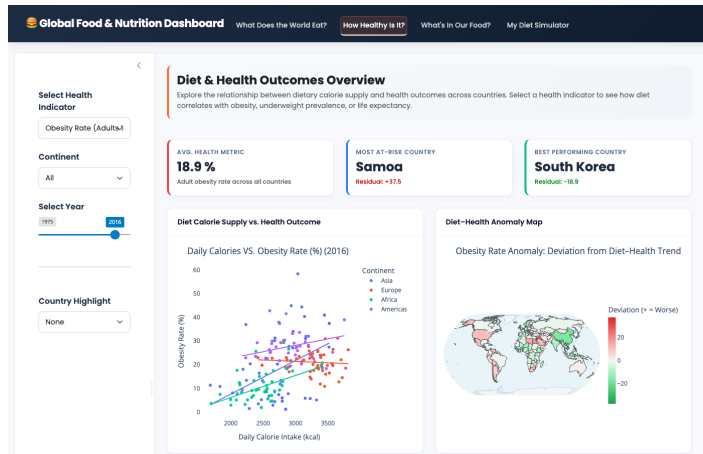
Supervised Learning (Product-Level)

Random Forest Classifier: Predicts Nutri-Score grades in real-time based on interactive macronutrient inputs (fat, sugar, salt, etc.).

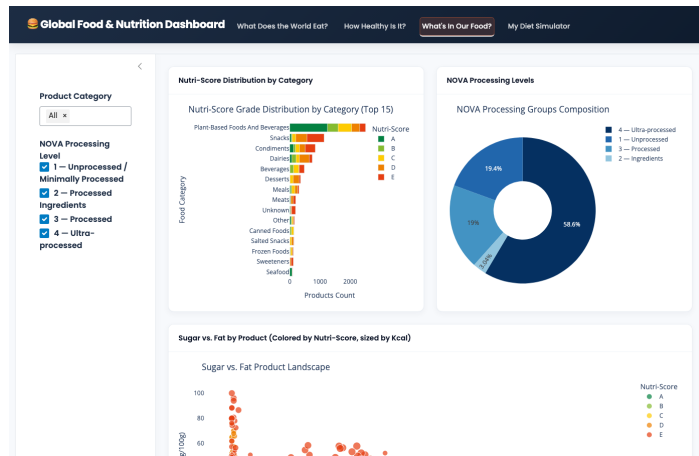
What does the world eat?



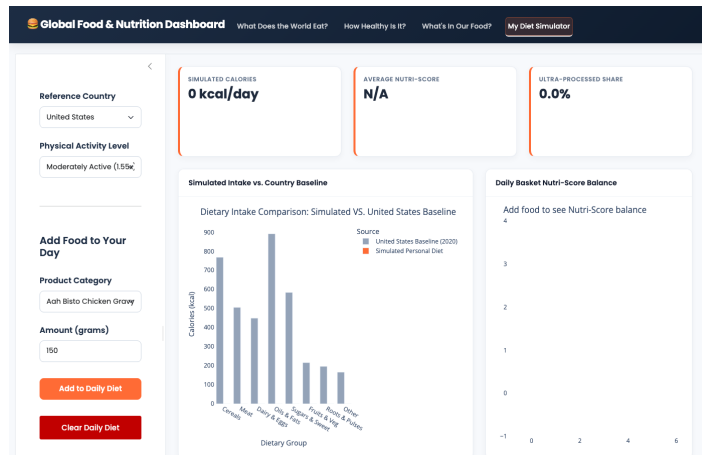
How healthy is it?



What's in our food?



My Diet Simulator



Findings & Lessons Learnt

Key Findings:

- High-calorie diets correlate strongly with obesity, but **composition** (sugar/fat ratios) is a stronger predictor of health anomalies than total supply alone.
- Ultra-processed foods dominate specific categories, heavily impacting average Nutri-Scores.

Lessons Learnt:

- *Challenge:* Managing Shiny's reactive graph to keep the app performant with tens of thousands of data points.
- *Solution:* Effective use of reactive values and selective UI rendering.

Q & A

Thank You!

Any Questions?

GitHub Repository:

<https://github.com/miinguyen/global-food-nutrition-dashboard>